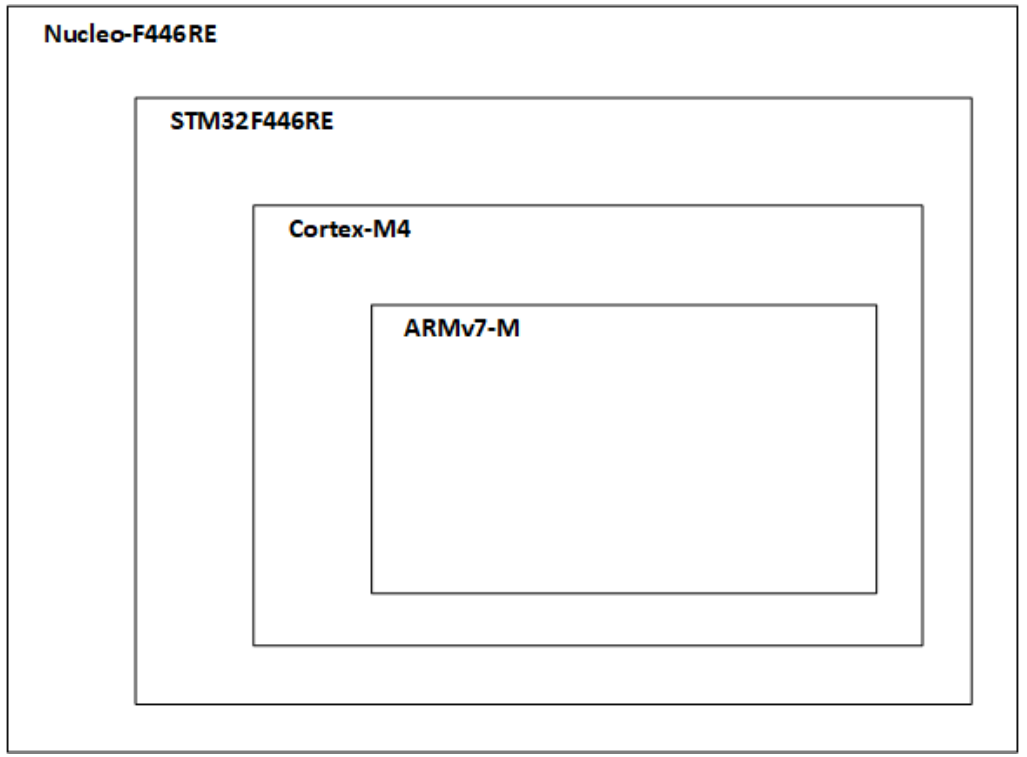
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Introduction to STM32F446RE



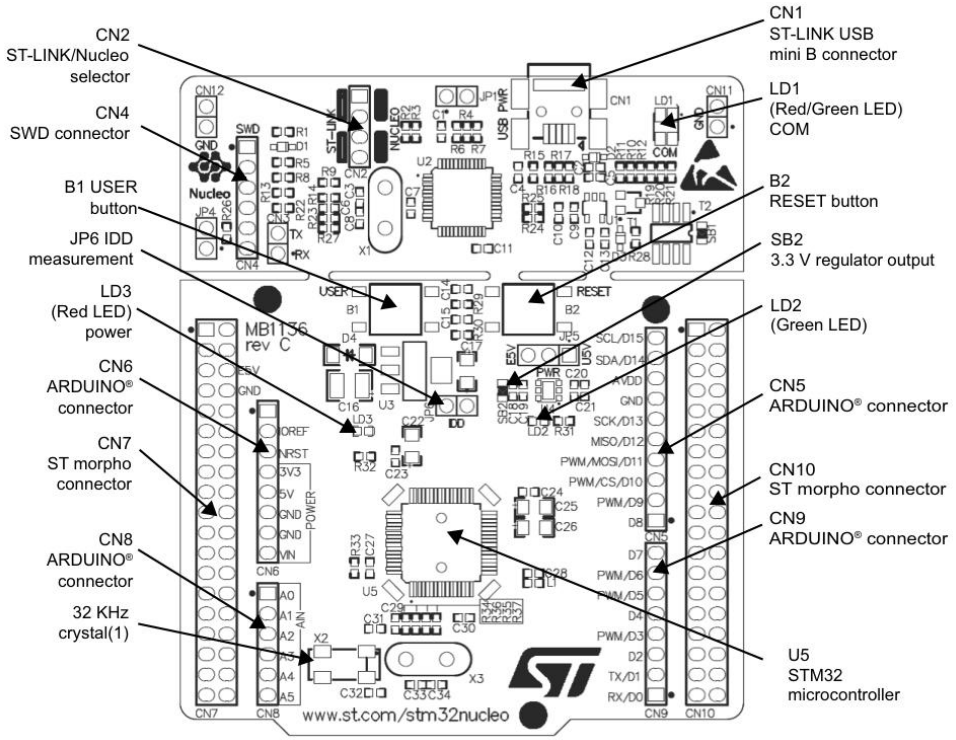
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  - armv7-m
    - ARMv7 Architecture Reference Manual.pdf
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  - cortex-m4
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    - Cortex-M4 Processor - Technical Reference Manual.pdf
- stmicro
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    - PM0214 Programming Manual.pdf
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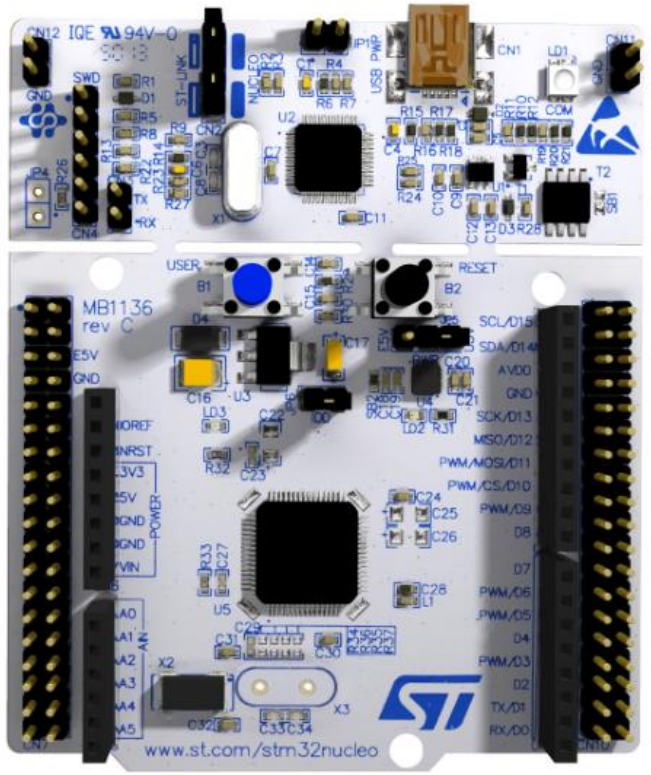
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Nucleo-64 STM32F446RE



Fair use, STMicroelectronics NV
UM1724 Rev 17, (STM32 Nucleo-64 boards, MB1136)

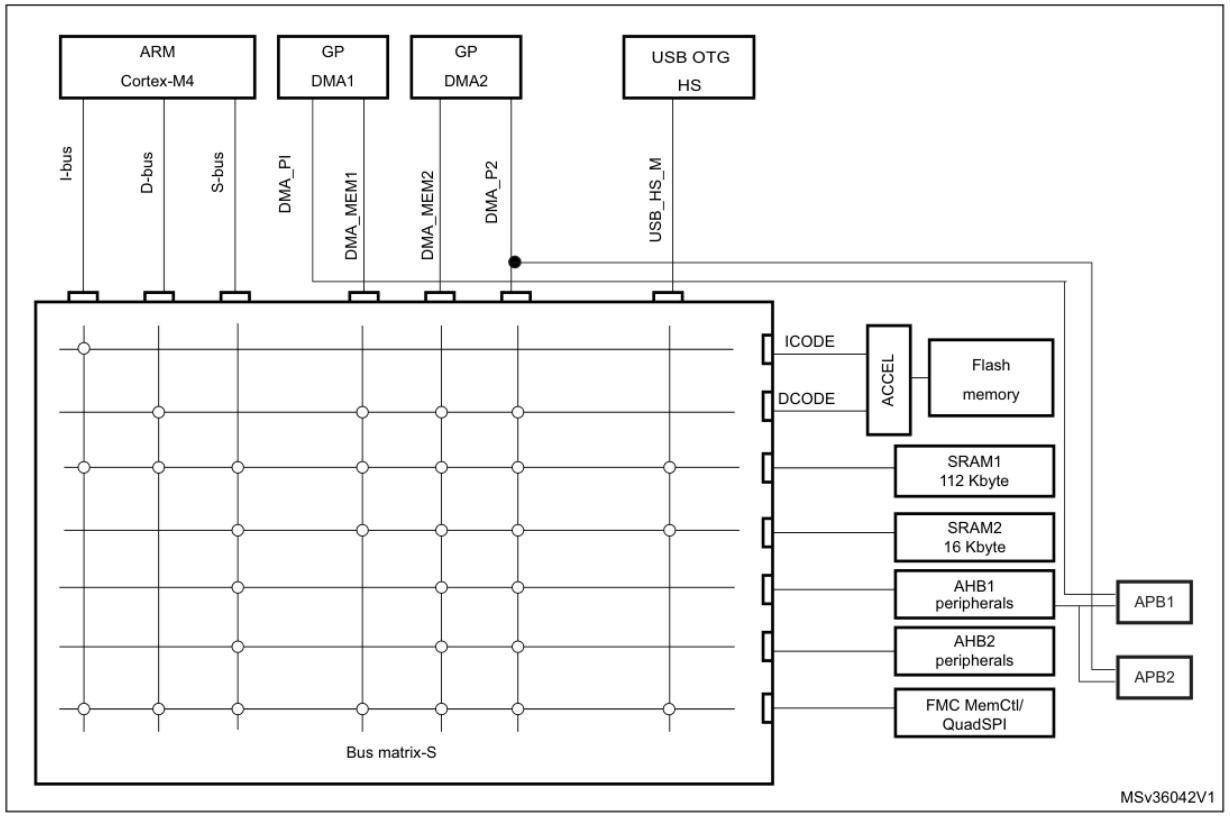
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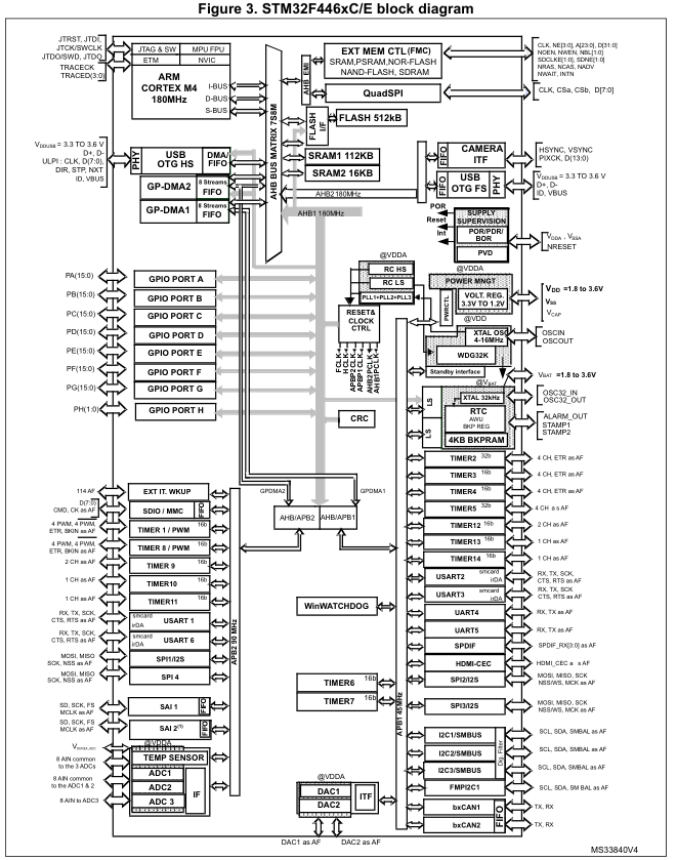
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Figure 1. System architecture for STM32F446xx devices



Fair use, STMicroelectronics NV
RM0390 Rev 9, (STM32F446xx advanced Arm-based 32-bit MCUs)

Figure 3. STM32F446xC/E block diagram



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STM32F446xC/E Datasheet DS10693 Rev 11

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Cortex-M4 Processor

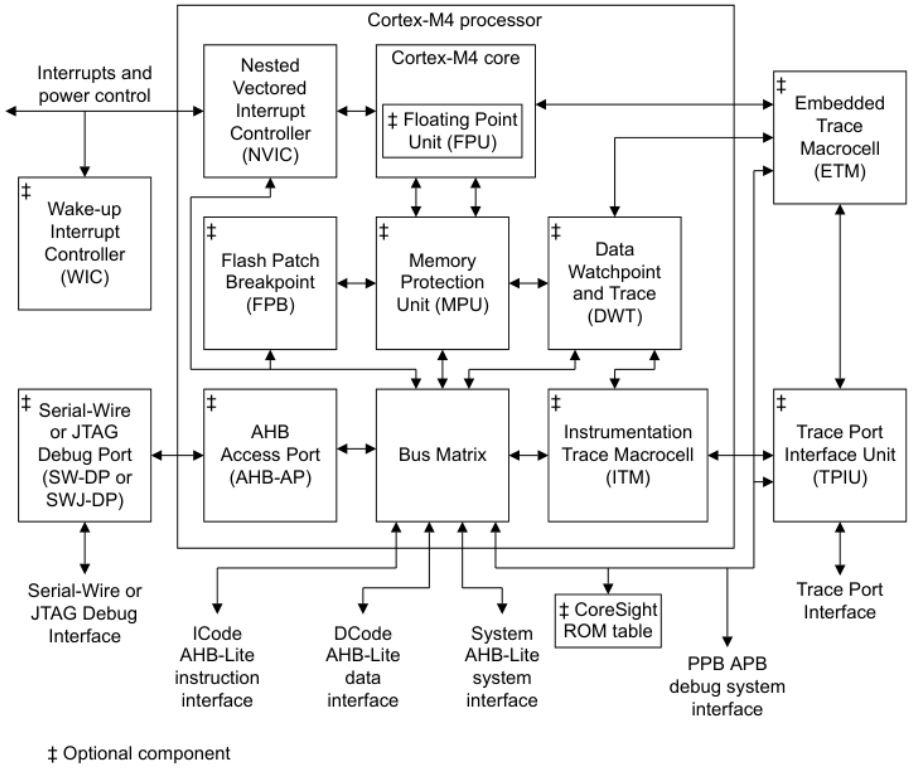


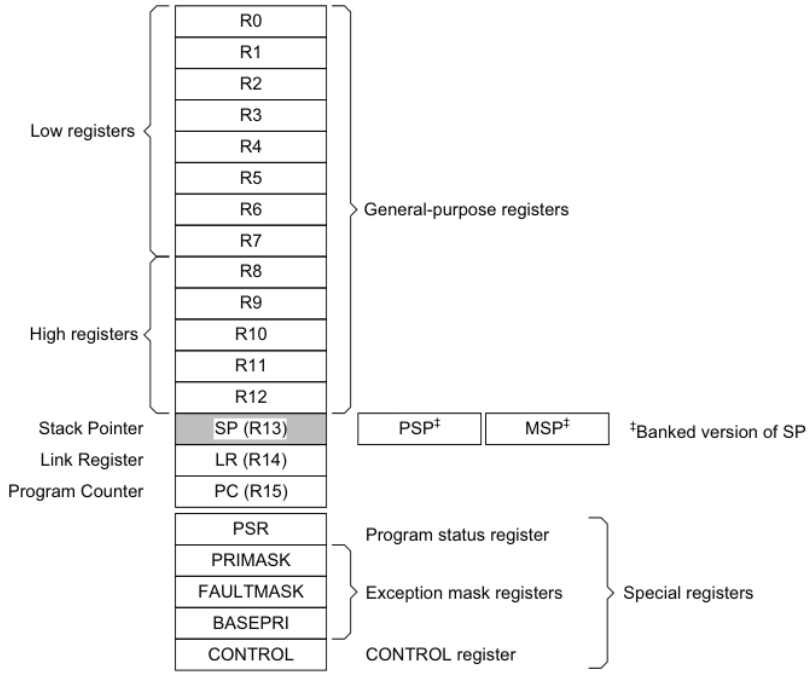
Figure 2-1 Cortex-M4 block diagram

Fair use, STMicroelectronics NV
Arm Cortex-M4 Processor
Technical Reference Manual (TRM), Revision r0p1

ARMv7-M Architecture + ISA

2.1.3 Core registers

The processor core registers are:



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Cortex-M4 Devices
Generic User Guide, Revision B Aug 2011

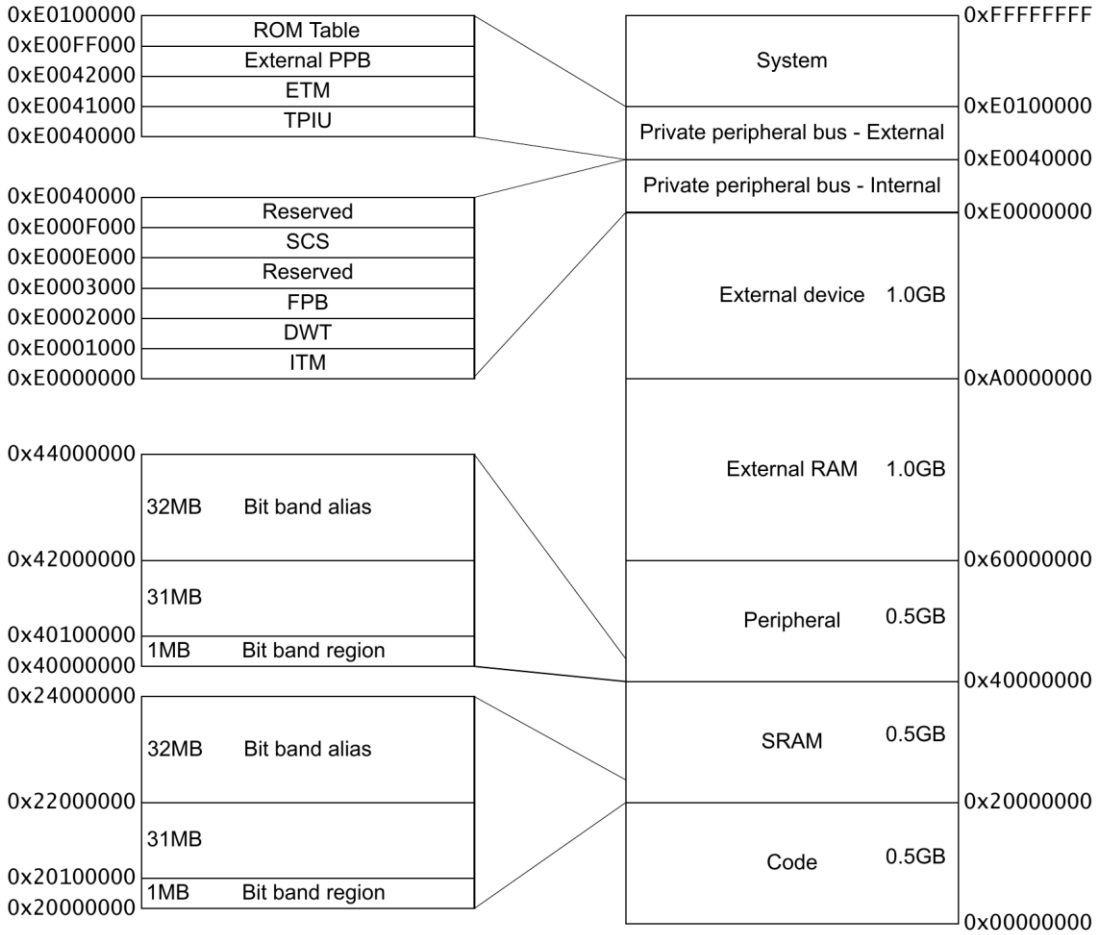
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Bit band region and alias (next slide)

ITM – Instrumentation Trace Macro
DWT – Data Watchpoint and Trace
FPB – Flash Patch and Breakpoint
TPIU – Trace Port Interface Unit
ETM – Embedded Trace Macrocell
PPB – Private Peripheral Bus
ROM Table – Capability Discovery Mechanism

SCS – System Control Space
- SCB – System Control Block
- NVIC – Nested Vector Interrupt Controller
- SysTick – System Tick Timer
- MPU – Memory Protection Unit
- FPU – Floating Point Unit

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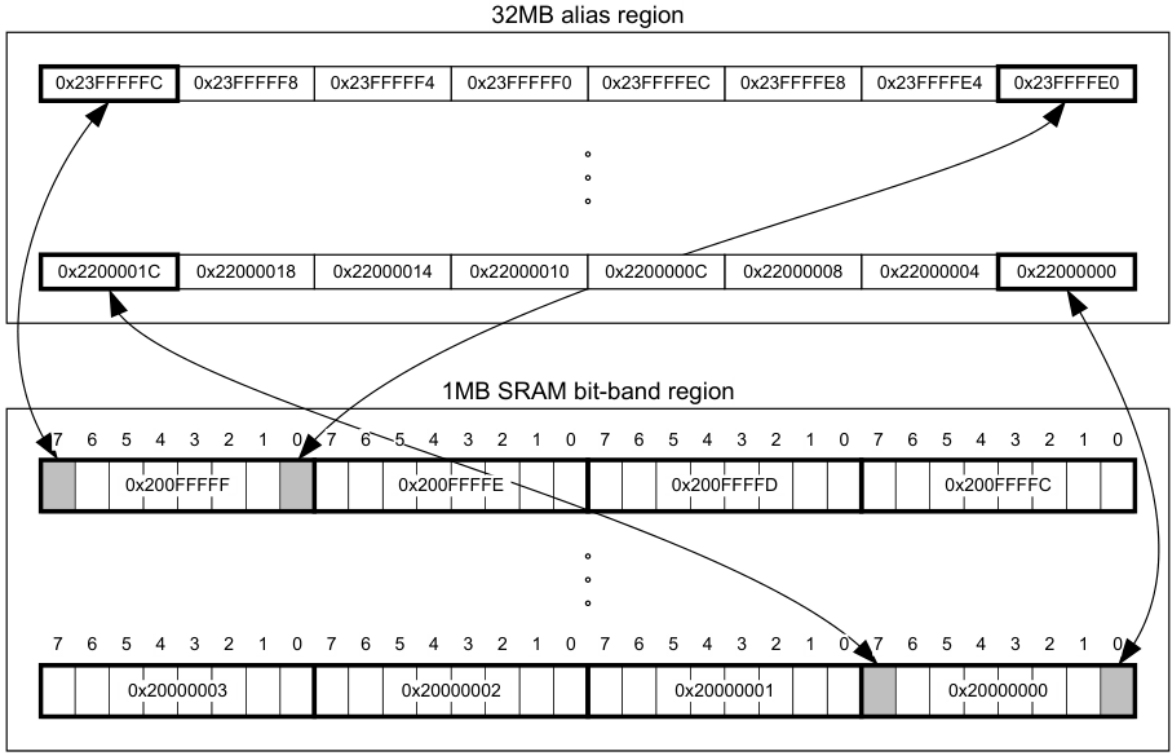


Figure 3-2 Bit-band mapping

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